

Strategic Practice 5

Section 1

1) 20 drops/min/in²

(area of 5 in² over t min
 ↳ 100t drops = $E(\text{raindrops on area over } t \text{ min}) = \lambda$

Poisson distribution b/c

- large # of events
- small probability of each event
- events are weakly dependent

$$\begin{aligned}
 P(\text{no raindrop on area in 3 seconds}) &= e^{-\lambda} \frac{\lambda^0}{0!} \\
 X=0 &= e^{-\lambda} \\
 &= e^{-(100 \cdot 0.05)} \\
 &= e^{-5} \\
 &\approx 0.0067
 \end{aligned}$$

2) a) 100 students, 2 courses

sitting randomly & independently assigned ^{for} each

$$\begin{aligned}
 &P(\text{no matches in seats}) \\
 &= 1 - P(\text{at least one seat match}) \\
 &= 1 - \frac{1}{100!} \sum_{n=1}^{100} (-1)^{n-1} \binom{100}{n} (100-n)! \\
 &= 1 - \frac{1}{100!} \sum_{n=1}^{100} (-1)^{n-1} \frac{100!}{n!(100-n)!} (100-n)! \\
 &= 1 - \sum_{n=1}^{100} (-1)^{n-1} \frac{1}{n!}
 \end{aligned}$$

b) WLOG assume student j is in seat j in course 1
 $j \in \{1, \dots, 100\}$.

$$\text{r.v. } I_j = \begin{cases} 0 & \text{if student } j \text{ isn't in seat } j \text{ for course 2} \\ 1 & \text{if " " is " " " " " " " "} \end{cases}$$

$$E(\# \text{ of students in }) = 100 \cdot \frac{1}{100} = 1 \quad \text{by symmetry \&}$$

matching seats

linearity

Poisson Approximation $\rightarrow \lambda = 1$

$$P(\text{matching seat} = 0) = e^{-1} \frac{1^0}{0!} = 1/e$$

$$\begin{aligned} c) \quad & P(\text{at least 2 matching seats}) \\ &= 1 - P(\emptyset \text{ matching seats}) - P(\text{exactly 1 matching seat}) \\ &= 1 - 1/e - e^{-1} \frac{1^1}{1!} \\ &= 1 - 2/e \end{aligned}$$

$$3) \quad X \sim \text{POIS}(\lambda) \quad \theta = e^{-3\lambda}$$

to estimate θ , we have an estimator for $g(X)$

bias of $g(X)$ is $E(g(X)) - \theta$

↳ if 0, then estimator is unbiased

$$a) \quad T = e^{-3X}$$

$$\begin{aligned} E(T) - \theta &= E(T) - e^{-3\lambda} \\ &= \sum_{k=0}^{\infty} e^{-3k} e^{-\lambda} \frac{\lambda^k}{k!} - e^{-3\lambda} \quad \text{by LOTUS} \\ &= \sum_{k=0}^{\infty} \frac{(e^{-3}\lambda)^k}{k!} e^{-\lambda} - e^{-3\lambda} \\ &= e^{-\lambda} e^{(e^{-3}\lambda)} - e^{-3\lambda} \end{aligned}$$

b) show $g(X) = (-2)^X$ is an unbiased estimator

$$\begin{aligned} E(g(X)) - \theta &= \sum_{k=0}^{\infty} (-2)^k e^{-\lambda} \frac{\lambda^k}{k!} - e^{-3\lambda} \\ &= \sum_{k=0}^{\infty} \frac{(-2\lambda)^k}{k!} e^{-\lambda} - e^{-3\lambda} \\ &= e^{-\lambda} e^{-2\lambda} - e^{-3\lambda} = 0 \end{aligned}$$

- c) $g(X)$ is a 'silly' estimator b/c when k is odd $(-2)^k$ is negative. The support of $\text{Pois}(\lambda)$ is non-negative and so are the values of its PMF

Section 2

1) $Z \sim N(0, 1)$ $S : \begin{matrix} 1 & 1/2 \text{ prob} \\ -1 & 1/2 \text{ prob} \end{matrix}$
 $P(Z \leq z) = \Phi(z)$

PDF of Z : $f(z) = c e^{-z^2/2}$

show $SZ \sim N(0, 1)$

$$\begin{aligned} P(SZ \leq z) &= P(SZ \leq z | S=1) P(S=1) \\ &\quad + P(SZ \leq z | S=-1) P(S=-1) \\ &= P(SZ \leq z | S=1) \frac{1}{2} + P(SZ \leq z | S=-1) \frac{1}{2} \\ &= P(Z \leq z) \frac{1}{2} + P(-Z \leq z) \frac{1}{2} \\ &= \Phi(z) \frac{1}{2} + P(Z \geq -z) \frac{1}{2} \end{aligned}$$

by symmetry {

$$\begin{aligned} &\downarrow P(Z \leq z) \frac{1}{2} \rightarrow \text{How?} \\ &P(Z \geq -z) = 1 - P(Z < -z) \\ &= 1 - \Phi(-z) \\ &= \Phi(z) \\ &\downarrow = \Phi(z) \frac{1}{2} + \Phi(z) \frac{1}{2} = \Phi(z) \end{aligned}$$

so $SZ \sim N(0, 1) \rightarrow$ has same CDF as $N(0, 1)$

- 2) $P(X < Y) = P(Y < X)$ if X and Y are i.i.d. by symmetry. $P(X < Y) = P(Y < X) = 1/2$ since its equally likely $X < Y$ and $Y < X$. If X and Y were not independent then $P(X < Y)$ might not equal $P(Y < X)$, since X gives information about Y that might impact X vs. Y (and vice versa)

↙ discrete case $P(X < Y) = P(Y < X)$ but $P(X = Y) > 0$
so $P(X < Y) < \frac{1}{2}$
continuous case $P(X < Y) = P(Y < X) = \frac{1}{2}$ since
 $P(X = Y) = 0$

3) $X \sim \text{Bin}(n, p)$, explain $n - X \sim \text{Bin}(n, 1 - p)$

This makes sense. In the former, you're looking at the distribution of # of successes in n $\text{Bern}(p)$ trials. In the latter, it's the dist. of # of failures of n $\text{Bern}(p)$ trials.

4) I've done this problem before!

Key observation: of the passengers who get the wrong seat, there's only one way they can be ordered. Let's say passengers $1, X_1, \dots, X_n$ have shuffled seats. $1 < X_1 < \dots < X_n$. The only way/ order this can happen is passenger 1 takes seat X_1 ; passenger X_1 takes seat $X_2, \dots$, passenger X_n takes seat 1.

Therefore seat 100 (last seat) can only be occupied by passengers 1 or 100. These seats are symmetric so the probability seat 100 is occupied by passenger 100 is $\frac{1}{2}$.

Section 3

1) $Y = e^X$ $X \sim N(\mu, \sigma^2)$
↳ "logNormal" distribution

$$\begin{aligned}
 \text{CDF } Y: P(Y \leq y) &= P(X \leq \ln(y)) && \text{ln of each side} \\
 &= P\left(\frac{X-\mu}{\sigma} \leq \frac{\ln(y)-\mu}{\sigma}\right) && \text{standardize} \\
 &= \Phi\left(\frac{\ln(y)-\mu}{\sigma}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{PDF } Y: & \text{ take derivative of CDF} \\
 &= \frac{1}{y\sigma} \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{\ln(y)-\mu}{\sigma}\right)^2/2} && \text{by chain rule}
 \end{aligned}$$

2) $U \sim \text{Unif}(0,1)$ construct r.v. X whose PDF is $\lambda e^{-\lambda x}$ for $x > 0$; $\lambda > 0$ & constant

$$\begin{aligned}
 \text{CDF: } P(X \leq x) &= \int_0^x \lambda e^{-\lambda t} dt = \lambda \int_0^x e^{-\lambda t} dt \\
 &= -e^{-\lambda t} \Big|_0^x = -e^{-\lambda x} + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{compute inverse} \quad y &= 1 - e^{-\lambda x} \\
 1 - y &= e^{-\lambda x} \\
 \ln(1-y) &= -\lambda x \\
 x &= -\frac{1}{\lambda} \ln(1-y)
 \end{aligned}$$

$$F^{-1}(U) = -\frac{1}{\lambda} \ln(1-U) \sim X \quad \text{by universality of uniform thm.}$$

3) $Z \sim N(0,1)$ find $E(\Phi(Z))$ w/o LOTUS
where Φ is CDF of Z

$$\text{CDF } Z: F(X \leq x) = \Phi(x)$$

By the universality of uniform theorem:

$$Z \sim \Phi(x) \rightarrow Z = \Phi^{-1}(U)$$

$$\therefore \Phi(Z) = \Phi(\Phi^{-1}(U)) = U$$

$$E(U) = \frac{0+1}{2} = \frac{1}{2}$$

4) $\text{Unif}(0, L)$ length of first piece, stick has length L

r.v. X — length of larger piece

CDF of X :

$$F = \int_{-\infty}^x f(t) dt = \begin{cases} 0 & x < \frac{1}{2}L \\ \frac{2}{L}(x - \frac{1}{2}L) & \frac{1}{2}L \leq x \leq L \\ 1 & x > L \end{cases}$$

$$X \sim \text{Unif}(\frac{1}{2}L, L) \quad E(X) = \frac{3}{4}L$$

Section 4

1) $X \sim \text{Pois}(\lambda)$, find $E(X!)$ by LOTUS

$$\begin{aligned} E(X!) &= \sum_{k=0}^{\infty} k! e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \lambda^k \quad \text{geometric sum} \\ &= e^{-\lambda} \frac{1}{1-\lambda} \quad \text{for } 0 < \lambda < 1 \end{aligned}$$

2) $Z \sim N(0, 1)$. Find $E|Z|$

$$\begin{aligned} E(Z) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} |z| e^{-z^2/2} dz \\ &= \int_0^{\infty} \frac{1}{\sqrt{2\pi}} z e^{-z^2/2} dz \\ &= \frac{2}{\sqrt{2\pi}} (-e^{-z^2/2}) \Big|_0^{\infty} = \sqrt{\frac{2}{\pi}} \end{aligned}$$

3) $X \sim \text{Geom}(p)$ t is a constant
Find $E(e^{tX})$

$$E(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} p(1-p)^k$$

$$= p \sum_{k=0}^{\infty} (e^t q)^k$$

$$= p \frac{1}{1-e^t q} \quad 0 < e^t q < 1$$